



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to Sri Hari Agencies

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

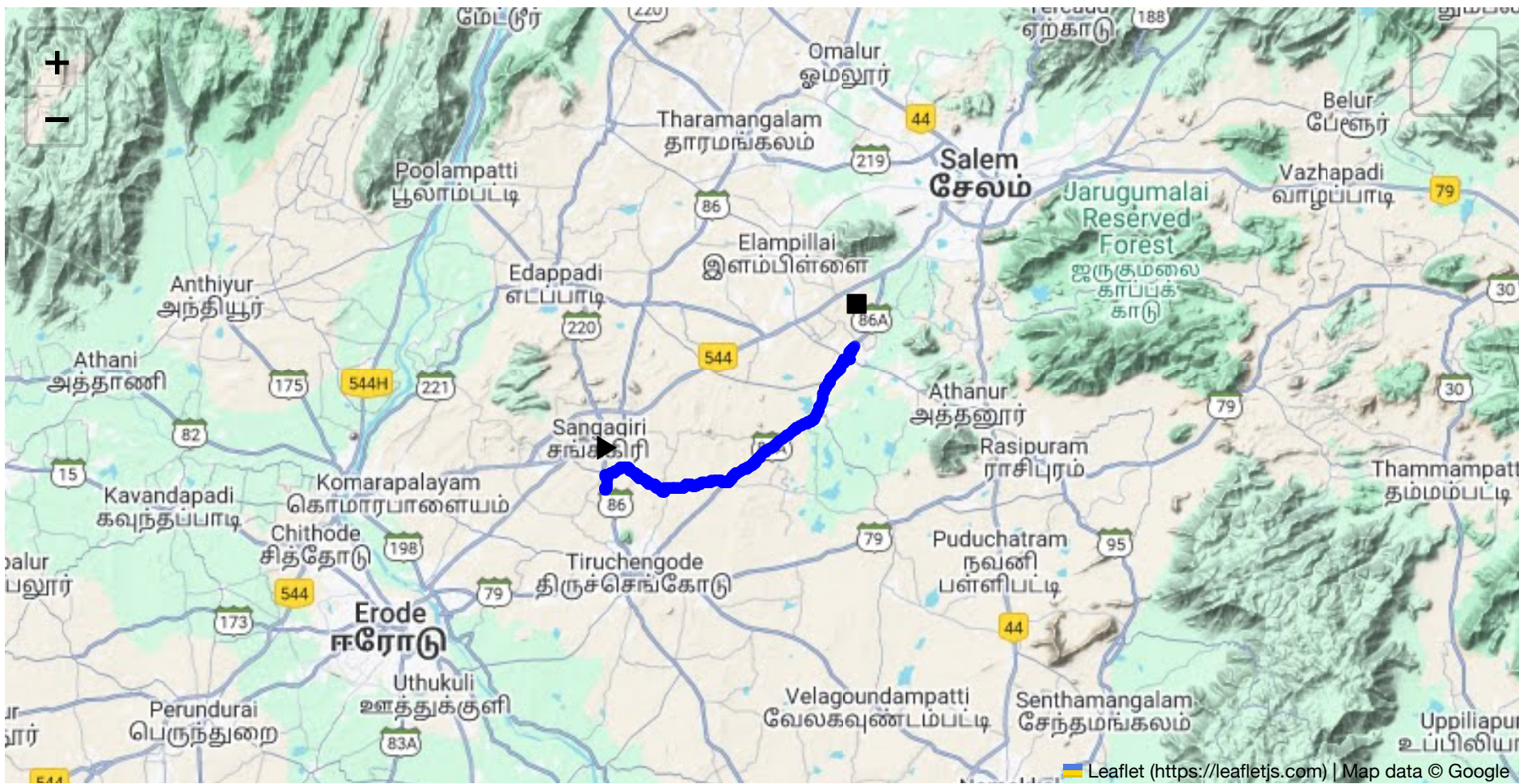
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 28.10 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours

Start: (11.4381, 77.8734)
End: (11.540842, 78.054573)



Welcome to the Journey Risk Management Study

Route Analysis Overview

1. Route Map Overview
- The route begins in Sangagiri, Tamil Nadu, and proceeds through Morepalayam and Kammalankadu before reaching Attayampatti, Salem. The journey generally involves local roads and possibly state highways, spanning approximately 28.10 km.
2. Typical Weather Conditions and Potential Weather-Related Hazards
- The region experiences tropical weather conditions. Summers are usually hot and dry, while monsoons (from June to September) bring heavy rainfall, potentially leading to road flooding, especially on poorly maintained sections. Fog can be an issue during the early winter mornings.
3. Traffic Patterns
- Peak traffic hours are typically in the morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) as local commuters travel to and from work.
 - Congestion might be encountered in smaller villages and at intersections with major roads near urban centers like Salem.
4. Road Quality and Infrastructure Assessment
- Road quality is varied; isolated stretches might be in poorer condition, especially in rural segments. Careful driving is required due to potential potholes and uneven surfaces.
 - Some sections may lack adequate signage, particularly in rural areas.
5. Alternative Routes for Emergencies
- In case of road closures or emergencies, drivers can take bypass routes through nearby towns like Edappadi or interior village roads, though these may not support heavy vehicles well.
6. Local Regulations Affecting Hazardous Material Transport

- Regulations include restrictions on the movement of hazardous materials during night hours and through certain densely populated areas.
- Mandatory documentation and appropriate vehicle placarding are required to transport hazardous materials.

7. **Historical Incidents Involving Heavy Vehicles or Hazardous Materials**

- There have been isolated incidents of vehicle accidents due to poor road conditions or negligence, although no major hazardous material spills have been frequently reported.
- Monitoring local news channels for incident reports during transit is advisable.

8. **Environmental Considerations and Sensitive Areas**

- Routes near water bodies or agricultural fields must be navigated with extra caution to prevent contamination in the event of an accidental spill.
- Awareness of local wildlife crossings, especially in rural and forest-adjacent areas, is crucial.

9. **Communication Coverage Analysis**

- Generally, there is comprehensive mobile coverage throughout the route, but localized dead zones may exist in rural stretches and near forested areas.
- Reliable communication devices should be carried in case of emergencies.

10. **Estimated Emergency Response Times**

- Within urban and semi-urban areas, emergency response times are typically swift (15-30 minutes).
- In rural or isolated areas, response times could extend beyond 45 minutes to an hour.

11. **Overall Summary of Risk Assessment**

- The route presents standard risks encountered in rural and transitioning urban areas in Tamil Nadu. Risks include potholes, potential flooding in rainy seasons, congestion in market towns, and regulatory compliance checks for hazardous materials.
- Planning for weather variations, understanding local traffic patterns, staying informed of road conditions, and preparedness for communication challenges serve to mitigate these risks effectively.

In conclusion, while traversing this route with hazardous materials, adherence to speed regulations and situational awareness of road and weather conditions will significantly enhance safety and reduce potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	High	11.43673, 77.91703	15 KM/Hr
6	Turn	High	11.43926, 77.91705	15 KM/Hr
7	Turn	Medium	11.44030, 77.93066	30 KM/Hr
8	Turn	Medium	11.44075, 77.93216	30 KM/Hr
9	Turn	Medium	11.44088, 77.93223	30 KM/Hr
10	Turn	Medium	11.44127, 77.93218	30 KM/Hr
11	Turn	Medium	11.44243, 77.93244	30 KM/Hr
12	Turn	Medium	11.44254, 77.93256	30 KM/Hr
13	Turn	Medium	11.44550, 77.95329	30 KM/Hr
14	Turn	Medium	11.44369, 77.96036	30 KM/Hr
15	Turn	High	11.44387, 77.96283	15 KM/Hr
16	Turn	Medium	11.53064, 78.04684	30 KM/Hr
17	Turn	Medium	11.53170, 78.05073	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	police	Mallasamudram Police Station	11.4998427, 78.0303028	30 km/h	Medium
6	police	Attayampati C 1 Police Station	11.5313534, 78.050867	30 km/h	Medium
7	hospital	Mythili Hospital	11.533594, 78.050881	30 km/h	Medium
8	hospital	Dhanalakshmi Hospital	11.5318866, 78.0514614	30 km/h	Medium
9	hospital	Malligai Nursing Home	11.5308602, 78.0529837	30 km/h	Medium
10	clinic	A. Thangavel Physiotherapy Clinic	11.5314071, 78.052862	30 km/h	Medium
11	hospital	Nalam Hospital	11.5315309, 78.0533249	30 km/h	Medium
12	hospital	Sree Vasantham Hospital	11.540146, 78.0545787	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium
4	college	Mahendra Arts and science college	11.5112305, 78.0344287	30 km/h	Medium
5	school	Mahendra Matriculation School	11.5120354, 78.0354797	30 km/h	Medium
13	school	Vivekananda Balamandir Matric School	11.5395679, 78.0559776	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43673, 77.91703



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44075, 77.93216



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44088, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44127, 77.93218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44254, 77.93256



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44550, 77.95329



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44369, 77.96036



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44387, 77.96283



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53064, 78.04684



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53170, 78.05073